

Can Word Embeddings Handle Words With Multiple Meanings?

GloVe vs BERT on WiC

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INTRODUCTION

What is the problem?

The same word can mean different things. **Bank** can describe a river edge or a financial institution. This is **lexical ambiguity**. A representation that stays identical everywhere cannot separate those uses.

BACKGROUND

What do the terms mean?

Homonymy describes unrelated meanings. **Polysemy** describes related meanings. An **embedding** is a word represented as a vector. **Cosine similarity** measures how closely two vectors point in the same direction. GloVe is **static**: one vector per word [2]. BERT is **contextual**: the target vector changes with the sentence [3].

RELATED WORK

What have others done?

Word sense disambiguation first relied on dictionaries such as WordNet. Multi-sense embeddings then assigned several vectors to one word [4]. ELMo and BERT build vectors from context. WiC asks whether a word keeps the same meaning in two sentences without requiring a named dictionary sense [1].

HYPOTHESIS

What did we expect?

The GloVe target diagnostic should remain at chance. Nearby GloVe words should help slightly. Frozen BERT should be clearly better because it encodes each target in its sentence.

METHODOLOGY

What did we do?

We used 5,428 WiC training pairs and 638 separate validation pairs. The 1,400 public test pairs have no public labels, so we report no test score. For each pair we compared cosine similarity using three representations: GloVe target only, GloVe nearby-word average at ± 2 , and frozen bert-base-uncased target vectors averaged over the last four layers. Settings and thresholds were selected on training data only. Validation was used once. Intervals use 1,000 paired bootstrap resamples.

ERROR ANALYSIS

Where do they fail?

BERT alone solves 178 pairs and GloVe context alone solves 104. Both are wrong on 106 pairs. BERT finds 77.4% of same-sense pairs but 56.7% of different-sense pairs. GloVe context finds 46.4% and 64.6%. BERT makes 210 errors, including “engrave a letter” versus “engrave a pen.”

CONCLUSION

So what?

Context changes the answer. Frozen BERT separates meanings better than local GloVe context, but 67.1% is not a solved task. This result concerns one English benchmark, one frozen BERT model and a simple static baseline.

RESULTS

What did we find?

50.0%

GloVe target
chance diagnostic

55.5%

GloVe context
nearby words

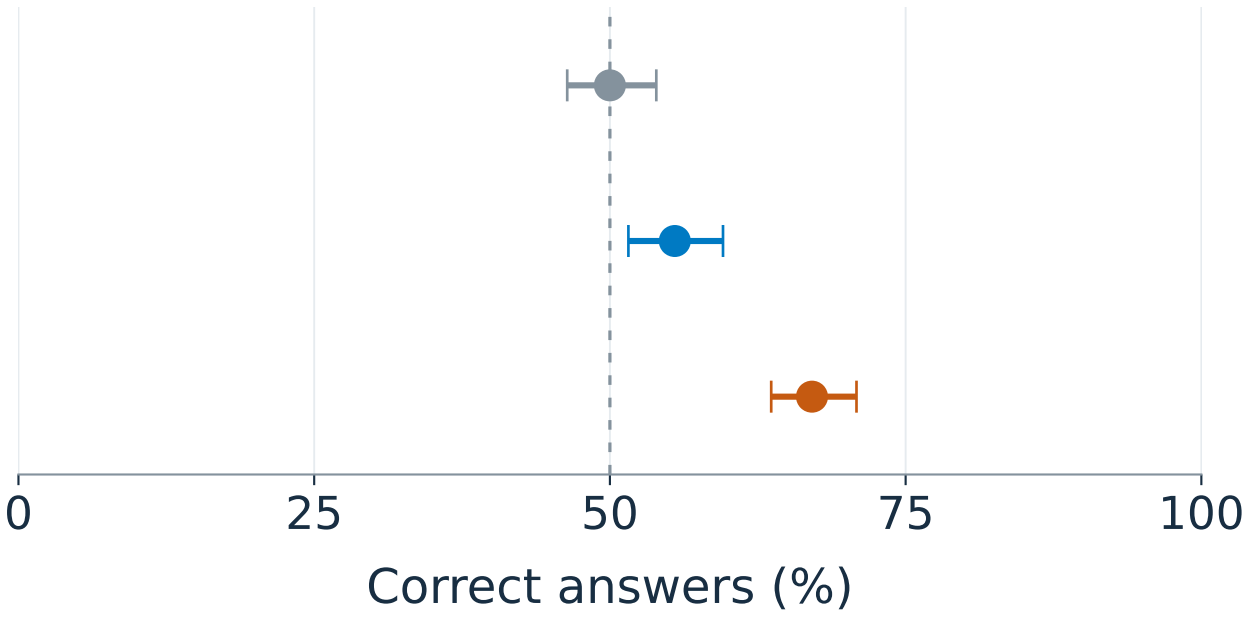
67.1%

BERT target
sentence-aware

GloVe: word only
One fixed vector for the target word

GloVe: nearby words
Average of nearby word vectors

BERT: word in its sentence
Target vector changes with context



50.0%
319 / 638 correct

55.5%
354 / 638 correct

67.1%
428 / 638 correct

+11.6 points

BERT over GloVe context. Paired 95% interval: 6.6 to 16.9 points. Layer 9 reaches 66.6%, compared with 63.6% at layer 12.

References

- [1] Pilehvar & Camacho-Collados (2019). *WiC*. NAACL.
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[6] Haber & Poesio (2021). *Polysemy and Homonymy in Contextualised Language Models*. Findings of EMNLP.